

Chapter

Water Resources

1. Which means of irrigation can be developed with minimum expenditure?

MAIN 2024

- (a) Tube well
- (b) Canal
- (c) Well
- (d) Drip irrigation

2. Which of the following is a ground water source? COMP 2023

- (a) Pond
- (b) Lake
- (c) Spring
- (d) River

3. A planned protection, improvement and the wise use of natural resources is known as: MAIN 2023

- (a) Degradation
- (b) Erosion
- (c) Depletion
- (d) Conservation

4. The Bamboo Drip system is common in: COMP 2023

- (a) Western states
- (b) Eastern states
- (c) North-Eastern states
- (d) Southern states

5. Why is there a need to conserve water in India? SQP 2021

- (a) A lot of water is polluted
- (b) No water is found below the land in India
- (c) Water is in abundance in India
- (d) We receive lot of rain in summer

6. What are the advantages of rain water harvesting? COMP 2021

- (a) Recharge of ground water and conservation of water
- (b) Provide water through canals to the farms and its conservation
- (c) It helps in harvesting of crops and its storage
- (d) Provide water for multinational companies to produce packaged drinking water

7. Why is canal irrigation more popular in Northern Plains? MAIN 2021

- (a) The surface of land is almost flat and Soft soil is easy to dig
- (b) To supply water to a number of cities in the plain area
- (c) Rocky surface of the plain help in their construction
- (d) Rivers are seasonal and so canals are needed

8. Tank irrigation is popular in: SQP 2022

- (a) South India
- (b) North West India
- (c) Eastern India
- (d) Coastal region

9. What is the advantage of using Drip irrigation? MAIN 2024

10. Why is Drip irrigation not practiced on a large scale in India? COMP 2024

11. Where in India is Tank irrigation most widely used? SQP 2023

12. Give one factor that has led to its popularity in the region mentioned by you. MAIN 2023

13. What are the sources of freshwater ? MAIN 2022

14. Name any two social movements which have been launched against the multi-purpose projects. COMP 2023

15. Name any two states which are involved in the Krishna-Godavari-dispute. MAIN 2023

16. What is a Tank or tanka ? MAIN 2022

17. Mention one advantage of using tank as a means of irrigation. MAIN 2023

18. How people conserve or harvest water in hills and mountainous regions ? COMP 2023

19. What is the percentage of water on Earth? COMP 2022

20. Name any two traditional rainwater harvesting methods. SQP 2022

21. Give a reason to explain each of the following: MAIN 2023

(i) Sprinkler irrigation is gaining popularity in recent times.

(ii) Tube well irrigation is used in Punjab.

(iii) North India has more area under canal irrigation.

22. Mention one advantage and one disadvantage of using tank irrigation. MAIN 2024

23. Distinguish between Inundation canal and Perennial canal. MAIN 2023

24. Give a reason for each of the following: COMP 2023

(i) Agriculture sector in India needs the support of irrigation.

(ii) Well irrigation is a popular means of irrigation in Northern India.

25. Give two ways in which Rainwater Harvesting is important for India. MAIN 2023

26. What is Rainwater Harvesting? COMP 2023

27. What do you mean by groundwater resource? SQP 2023

28. Give two reasons why tanks are used for irrigation in the Deccan. MAIN 2023

29. Give a reason for the significance of irrigation in COMP 2023

(i) Punjab

(ii) Rajasthan

30. "Need of the hour is to conserve and manage our water resources." Mention any four reasons. SQP 2022

31. (i) What is meant by rainwater harvesting ?

(ii) State any two methods of rainwater harvesting ? MAIN 2023

32. (i) Name two methods of water harvesting in India.

(ii) Mention any two objectives of rainwater harvesting. COMP 2023

33. Give three reasons for conservation of water resources. MAIN 2022

34. The method by which water is stored in tanks during the rainy season and used when needed is: COMP 2022

(a) Well irrigation

(b) Sprinkler irrigation

(c) Rahat system

(d) Rainwater harvesting

Solution

1. (d) Drip irrigation

The means of irrigation that can be developed with minimum expenditure is **Drip irrigation**.

2. (c) Spring

A **spring** is a ground water source.

3. (d) Conservation

A planned protection, improvement, and wise use of natural resources is known as **Conservation**.

4. (c) North-Eastern states

The Bamboo Drip system is common in **North-Eastern states** of India.

5. (a) A lot of water is polluted

There is a need to conserve water in India because **a lot of water is polluted**.

6. (a) Recharge of ground water and conservation of water

The advantages of rainwater harvesting are **recharging groundwater and conserving water**.

7. (a) The surface of land is almost flat and soft soil is easy to dig

Canal irrigation is more popular in the Northern Plains because **the land is almost flat and the soft soil is easy to dig**.

8. (a) South India

Tank irrigation is popular in **South India**.

9. The advantage of using **Drip irrigation** is that it **delivers water directly to the roots of plants**, ensuring **efficient water use**, minimizing wastage, and conserving water, especially in water-scarce regions.

10. Drip irrigation is not practiced on a large scale in India because it **requires a high initial investment** in equipment and setup, which many small-scale farmers cannot afford.

11. Tank irrigation is most widely used in **Southern India**, particularly in **Tamil Nadu and Andhra Pradesh**.

12. One factor that has led to the popularity of tank irrigation in Southern India is the **undulating terrain with natural depressions**, which makes it easy to **collect and store rainwater**.

13. Precipitation, surface run off and groundwater.

14. 'Narmada Bachao Andolan' and 'Tehri Dam Andolan.'

15. Karnataka and Andhra Pradesh.

16. A **Tank or Tanka** is a **man-made or natural structure** used for **storing and preserving rainwater**.

17. One advantage of using a tank for irrigation is that it **helps in water conservation** by storing rainwater, which can be used during **dry periods**, especially in regions with irregular rainfall.

18. In hills and mountainous regions, people conserve or harvest water by **building diversion channels** to redirect and collect water for use.

19. About **71% of the Earth's surface** is covered with water, of which **97.5% is saline** (found in oceans) and only **2.5% is freshwater**.

20. Two traditional rainwater harvesting methods are:

1. **Kuls** – used in Himachal Pradesh

2. **Khadins** – used in Gujarat

21. (i) **Sprinkler irrigation**: Gaining popularity because it **distributes water efficiently**, reduces wastage, and provides **even coverage**, especially in water-scarce areas.

(ii) **Tube well irrigation in Punjab**: Punjab relies on **groundwater**, and tube wells can irrigate large areas quickly and provide **year-round water supply**.

(iii) **Canal irrigation in North India:** Northern India has **permeable soil** and a **high groundwater level**, making it suitable for the development and efficiency of canal irrigation systems.

22. Advantage: Tank irrigation **stores rainwater**, making it available during **dry periods**, especially in regions with uneven rainfall.

Disadvantage: Tanks **occupy large areas of land**, reducing space available for agriculture and other uses.

23. Inundation Canal:

- It is a **seasonal canal** that carries water **only during the rainy season** when rivers are in flood.
- It **depends on excess floodwater** and may remain dry during other times of the year.

Perennial Canal:

- It is a **year-round canal** that carries water **throughout the year**.
- It is **fed by a dam, reservoir, or perennial river**, ensuring a continuous water supply for irrigation.

24. (i) Agriculture needs irrigation: Irrigation is essential in India because **rainfall is irregular and seasonal**, so irrigation ensures that crops **receive water consistently**.

(ii) **Well irrigation in Northern India:** Wells are popular because the region has a **high water table**, making it easy to access groundwater for irrigation.

25. Two ways in which Rainwater Harvesting is important for India:

(i) **Reduces dependency on groundwater:** It **replenishes groundwater levels**, which are declining in many regions.

(ii) **Mitigates water scarcity:** It provides a **sustainable water source** in areas with irregular or insufficient rainfall, especially during dry seasons.

26. Rainwater Harvesting is the practice of **collecting and storing rainwater** from rooftops, land surfaces, or other catchment areas for **future use**, such as irrigation, domestic purposes, or **recharging groundwater**.

27. Groundwater resource refers to the **water found beneath the Earth's surface**, which **saturates the pores and fractures** in sand, gravel, and rock formations.

28. Two reasons why tanks are used for irrigation in the Deccan:

(i) **Undulating Terrain:** The uneven landscape makes canal construction difficult, so **tanks effectively capture and store rainwater**.

(ii) **Hard Rock Structure:** The rocky soil limits groundwater availability, making tanks an **efficient method to store surface water** for irrigation during dry periods.

29. (i) Punjab: Irrigation is significant because **water-intensive crops** like wheat and rice need a **consistent water supply** for high productivity.

(ii) **Rajasthan:** Irrigation is crucial because it **compensates for the arid climate and low rainfall**, allowing cultivation in desert and semi-arid areas.

30. Four reasons to conserve and manage water resources:

(i) To **protect against health risks**.

(ii) To **ensure food security**.

(iii) To **prevent degradation of natural ecosystems**.

(iv) To **safeguard future generations from water scarcity**.

31. (i) Rainwater Harvesting: It is the practice of **collecting and storing rainwater** from rooftops, land surfaces, or other catchment areas for **future use**, such as irrigation, domestic use, or **recharging groundwater**.

(ii) **Two methods of rainwater harvesting:**

1. **Roof-top rainwater harvesting:** Collecting rainwater from rooftops and storing it in tanks or reservoirs.

2. **Surface runoff harvesting:** Diverting rainwater from land surfaces into **ponds, tanks, or recharge pits** to store water or recharge groundwater.

32. (i) **Two methods of water harvesting in India:**

1. **Kuls** – used in Himachal Pradesh.
2. **Khadins** – used in Gujarat.

(ii) **Two objectives of rainwater harvesting:**

1. To **replenish groundwater** and reduce dependency on wells and tube wells.
2. To **provide a sustainable water source** in areas with irregular or insufficient rainfall.

33. Three reasons for conservation of water resources:

(i) **Health Protection:** Conservation helps reduce health risks from **contaminated water** polluted by urban waste, industrial effluents, and agricultural chemicals.

(ii) **Groundwater Preservation:** Overuse of groundwater lowers water tables, so conservation is needed to **restore balance**.

(iii) **Future Generations:** Saving water ensures that **future generations** have access to **clean and safe drinking water**.

34. Thus, the correct option is (d) **Rainwater harvesting**.